

The Coverage–Selection Gap in Tiny Recursive Models: Evidence from Augmentation Orbits

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We turned this project into a workshop paper and submitted it to the NewInML Workshop at NeurIPS 2026. We wrote the workshop paper using the workshop’s required template and format. This version is just a shorter version of the submitted paper, prepared for DL course project.

Abstract

Tiny Recursive Models (TRM) solve ARC-AGI-1 with a small recurrent network and a large test-time augmentation orbit. we ask whether the orbit contains correct answers that the standard whole-grid majority vote fails to select. On 400 public evaluation puzzles, a correct grid appears for 61.75% of puzzles under the $k = 1000$ candidate budget, while the B1 vote selects a correct grid for 40.00%, leaving a 21.75-point coverage–selection gap. Training-free selectors based on orbit structure do not close this gap, even though our initial thought was that it would. Orbit-level equivariance defect predicts held-out vote failure (AUROC 0.842), but candidate-level centrality often favors the same incorrect attractor as B1, the competing top candidates have a median of 96% cell agreement. Among the tested zero-training confidence signals, vote margin has the lowest AUC (0.2608). Blanking puzzle-identity embeddings reduces candidate coverage to 10.25%, indicating that identity information is used during generation. All reported values come from frozen cached artifacts and a committed 200/200 puzzle split.

1 Introduction

Tiny Recursive Models (TRM) reach strong accuracy on ARC-AGI with roughly seven million parameters and a recurrent computation scheme [Jolicoeur-Martineau, 2025, Chollet, 2019]. Their inference pipeline is also important, each puzzle is evaluated under a large set of augmentations, the predictions are mapped back to canonical coordinates, and a whole-grid vote selects one answer. This makes the final result depend on both the model and the aggregation rule.

We study a simple question: how often does the augmentation orbit contain a correct grid that the vote does not select? We call the first quantity *coverage* and the second *selection*. The distinction is familiar in repeated-sampling systems, where a correct answer can occur in the sample without being ranked first [Brown et al., 2024, Snell et al., 2024]. Here the candidates are generated by augmentation rather than independent language-model samples. We measure the gap, test several training-free selectors, and examine why orbit structure is useful for identifying difficult puzzles but less useful for choosing between competing whole-grid answers.

The analysis is transductive: the public checkpoint already contains the puzzle-identity embeddings used by the evaluation pipeline. Consequently, the held-out split measures the aggregation problem for a fixed candidate pool; it is not a test of model generalization to unseen identities.

2 Related Works

Recursive reasoners. Work following HRM and TRM has mainly changed training, test-time adaptation, sampling, or architecture [Wang et al., 2025, Asadulaev et al., 2026, McGovern, 2025, Sghaier et al., 2026, Wang and Reid, 2026, Gao et al., 2025]. Other analyses report shallow effective recursion, a strong role for identity conditioning, and convergence to confident incorrect attractors [Roya-Azar et al., 2025, Ren and Liu, 2026]. We keep the checkpoint and sampling procedure fixed and operate on its cached outputs.

Candidate selection. Answer-level voting, learned verifiers, process reward models, and product-of-experts rescaling are established ways to choose among multiple candidates [Wang et al., 2023, Cobbe et al., 2021, Lightman et al., 2024, Franzen et al., 2025]. Our selectors use only decoded grids and cached metadata; they do not perform another model pass.

Augmentation and abstention. Orbit averaging is a standard route to approximate invariance [Cohen and Welling, 2016], while selective prediction evaluates how accuracy changes as low-confidence cases are withheld [Geifman and El-Yaniv, 2017, Geifman et al., 2019]. We compare structural confidence with simple vote-derived baselines, which are often difficult to beat [Hendrycks and Gimpel, 2017].

3 Problem Definition and Method

Setup. We use the publicly released ARC Prize verification checkpoint of TRM. The upstream pipeline uses a 1,000-augmentation budget; when an augmentation group is exhausted, the effective number of emissions is recorded and the requested budget is clamped. Across 400 ARC-AGI-1 puzzles (419 pair-scoped descriptors), the cache contains 385,815 emissions and 92 invalid (0,0) grids. Invalid emissions remain in the emitted population and in the B1 vote map.

Predictions are canonicalized before analysis, and all reported selectors operate on this frozen cache with CPU-side computations.

Three audits gate the study: the online B1 score, a cache recomputation, and the published verification score all equal 40.00%; the dataset build reproduces 876,406 identifier-embedding rows; and inverse-transform round trips are bit-exact. The committed split contains 200 DEV puzzles (208 descriptors) and 200 TEST puzzles (211 descriptors) with no puzzle overlap. Hyperparameters were selected on DEV only, and TEST was evaluated once after the configuration was frozen.

For a task τ with canonicalized emissions c_i , we define the orbit-level equivariance defect as $\Delta(\tau) = 1 - \mathbb{E}_{i,j}[\text{sim}(c_i, c_j)]$. The pre-registered task-level hypothesis is that larger Δ predicts B1 failure; candidate-level selectors then test whether structure can rank a correct grid above the vote winner. The orbit is non-degenerate in aggregate: the puzzle-weighted means are 290.05 distinct grids, 38.67% modal vote mass, and 67.88% singleton-grid mass per orbit.

Coverage and selection. Coverage ($\text{pass}@k$) is the fraction of puzzles for which a correct grid appears among the candidates available at budget k . Selection ($\text{maj}@k$) is the fraction for which the aggregation rule ranks a correct grid first from the same candidate pool. B0 is the canonical identity prediction. B1 ranks distinct emitted grids by their emission count $n(c)$ and uses mean halt-head Q as the tie-breaker, matching the upstream evaluator.

M1: centrality weighting. Let $\text{sim}(c, c')$ be cell agreement when shapes match and zero otherwise. For a distinct grid c , define

$$d(c) = \text{mean}_{c' \in \mathcal{S}(c)} \text{sim}(c, c'), \quad w_i = \frac{\exp(\beta d(c_i))}{\sum_j \exp(\beta d(c_j))}.$$

The set or multiset $\mathcal{S}(c)$ is chosen in one of three implemented ways: *distinct* (other distinct grids), *multiset* (all emissions with their multiplicities), or *non_identical_support* (emissions excluding copies of c). Emission weights are accumulated by grid and then ranked with the same mean- Q tie-break as B1. At $\beta = 0$, all emissions have equal weight, so the grid scores reduce to emission counts and the ranking agrees with B1.

M2: cell-marginal scoring. M2 first chooses the modal output shape by weighted mass. Let I_m be the emissions with that shape and let $\tilde{w}_i = w_i / \sum_{j \in I_m} w_j$. The marginal distribution is

$$p_{r,s}(v) = \sum_{i \in I_m} \tilde{w}_i \mathbf{1}[y_i(r, s) = v].$$

For every *emitted* grid c of the modal shape, M2 computes

$$S_{M2}(c) = \lambda \sum_{r,s} \log \left((1 - \epsilon) p_{r,s}(y_{r,s}) + \frac{\epsilon}{10} \right) + (1 - \lambda) \log n_m(c).$$

where $n_m(c)$ is the count within the modal shape. No new grid is synthesized by cell-wise argmax. At $\lambda = 0$, M2 is vote-count ranking within the modal shape; it equals global

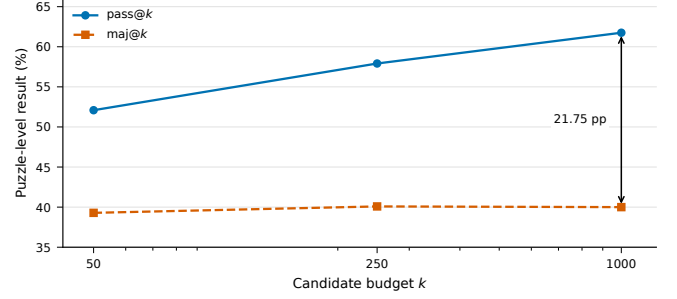


Figure 1: Coverage ($\text{pass}@k$) and B1 selection ($\text{maj}@k$) on the 400 ARC-AGI-1 evaluation puzzles. Coverage increases with the candidate budget, while selection remains close to 40%.

B1 only when the B1 winner has that shape. The frozen DEV settings are $\beta = 1$, $\lambda = 0.05$, $\epsilon = 10^{-6}$, and emitted marginal support.

The distinct-uniform marginal-support ablation was also evaluated on DEV and did not replace the emitted-support configuration.

M3 and Top-2. M3 screens candidates using support-derived shape relations. If the screen removes every candidate, the implementation falls back to the non-empty original set. The screen is active on 86% of held-out puzzles and changes no aggregate rank-1 result. For the committed two-attempt policy, the first answer is M2 rank-1 and the second is B1 rank-1; if they are identical, M2 rank-2 is used instead.

4 Implementation and Experimental Setup

We implemented a Python package with a NumPy-only analysis core for candidate schemas, similarity workspaces, aggregation, diagnostics, selective-prediction evaluation, caching, and shape screening, plus an adapter for the upstream TRM inference and canonicalization code. Inference was run once per dataset and its outputs were cached. Every reported number in this paper is a deterministic computation from those frozen artifacts. Paired confidence intervals use 10,000 puzzle-level bootstrap resamples [Efron and Tibshirani, 1994]. Figures and tables are generated from the frozen result tree so that the text and numerical artifacts share one source of truth.

For $k < 1000$, each budget uses ten shared random subsamples per descriptor (without replacement); all methods therefore use the same candidate subsets. The analysis seed is 20260807, and metrics average multiple test pairs within each puzzle before averaging puzzles.

5 Contributions and Results

(1) The coverage–selection gap. At $k = 50/250/1000$, coverage is 52.0875%/57.9125%/61.75%, while B1 selection is

Table 1: ARC-AGI-1 TEST (200 puzzles), frozen configurations and one evaluation. Values are percentages; deltas are percentage points.

Method	Rank-1	Top-2	Δ R1 vs B1
B0	29.25	29.25	-11.00
B1	40.25	45.50	+0.00
M1	40.25	45.50	+0.00
M1+M2	41.25	44.00	+1.00
M1+M2+M3	41.25	44.00	+1.00

39.2875%/40.0875%/40.00%. The corresponding gaps are 12.80/17.825/21.75 percentage points (Fig. 1). Thus, at the full budget, the cache contains a correct answer for substantially more puzzles than the vote answers correctly.

The compute-matched sweep does not establish a broad $4\times$ saving: across ten paired TEST subsamples, M1+M2 at $k = 250$ averages 40.50% Rank-1 versus 40.35% for B1 at $k = 1000$, but its Top-2 accuracy is 43.83% versus 45.50%.

(2) Structured selection is weak. Table 1 gives the frozen TEST comparison. M1 has the same held-out Rank-1 and Top-2 accuracy as B1, although its full rankings and MRR differ slightly. Adding M2 changes rank-1 accuracy from 40.25% to 41.25%, an exploratory +1.00-point difference with a paired bootstrap 95% interval of [0.00, 2.50] points and $p = 0.266$. Because the interval includes zero and the configuration was not selected on TEST, this is not evidence of a reliable general improvement; its MRR is also lower than B1 (0.4476 versus 0.4533). Under the committed two-attempt policy, Top-2 accuracy decreases from 45.50% for B1 to 44.00% for M1+M2 and M1+M2+M3. M3 adds no aggregate gain.

(3) Task-level signal versus candidate-level choice. Orbit-level equivariance defect predicts held-out B1 failure with AUROC 0.8419. The same diagnostic is 0.8279 on the pinned ARC2 supporting run and 0.7708 under ARC1 identity blanking, but neither result establishes transfer to a new checkpoint. On the 46 puzzle-equivalent covered DEV failures (47 pair descriptors), centrality prefers the correct alternative 14 times and the incorrect B1 winner 32 times. The leading alternatives have a median whole-grid similarity of 0.96, or 96% cell agreement, with a median disagreement fraction of 0.04. Shared background therefore dominates the geometry used by M1. The median empirical β at which a decision flips is 248.37, whereas the sweep reached only 16, so simply increasing the tested weight is not a supported repair. A discriminative-cell diagnostic improves local preference on DEV failures (21 correct-preferred versus 16 wrong-preferred puzzle cases, with 9 ties), but it lowers or fails to improve DEV selector accuracy and was not evaluated on TEST.

(4) Abstention. Structural defect is informative for selective prediction: its AURC is 0.3159 versus 0.5975 for random ordering. It is nevertheless not the best tested confidence signal. Vote margin, vote share, vote entropy, and winner mean- Q obtain AURCs of 0.2608, 0.2621, 0.2777, and 0.3128, respectively (Fig. 2). When puzzles are ranked by structural

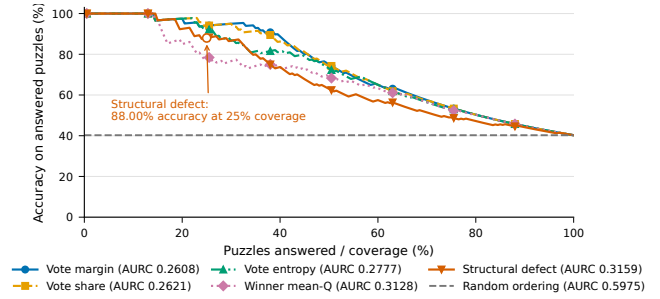


Figure 2: Risk-coverage curves on ARC-AGI-1 TEST. Lower AURC is better. Structural defect beats random ordering but is weaker than the vote-derived confidence signals.

defect, the most coherent quarter reaches 88.00% accuracy at 25% coverage, with a 95% bootstrap interval of [78%, 98%] and no additional inference.

(5) Identity acts at generation. Blanking the puzzle-identity embeddings reduces candidate coverage from 61.75% to 10.25% and held-out B1 rank-1 accuracy from 40.25% to 4.00% (95% CI [1.50%, 7.00%]). Across all retained blank-ID predictions, B1 is 3.5%. The historical upstream value of 0% is a scorer padding/masking artifact: identifier-zero rows are removed before voting. The retained-prediction ablation therefore indicates that identity conditioning affects candidate generation before aggregation, and that changing the selector cannot restore the missing coverage.

6 Conclusion and Future Work

For this checkpoint, augmentation reveals substantially more correct solutions than B1 ultimately selects, demonstrating that candidate selection, not only candidate generation, is an important remaining opportunity for improving TRM inference. Orbit structure provides a useful signal for identifying difficult puzzles, while our results show that whole-grid centrality alone is insufficient for distinguishing highly similar correct and incorrect attractors. M2 provides an encouraging exploratory rank-1 improvement, and vote margin offers a strong zero-training signal for selective prediction.

Under the computational constraints of a course-scale project, we focused on one frozen checkpoint and one inference configuration, using cached emissions to enable extensive and reproducible selector analysis without repeated GPU inference or checkpoint retraining. This design allowed controlled comparisons, while leaving several promising directions open. Future work could train a selector on label-complete training orbits, emphasize discriminative cells rather than background-dominated whole-grid similarity, allocate augmentation budgets adaptively using confidence signals, and combine disagreement across independently trained checkpoints. Evaluation on additional checkpoints and larger ARC2 runs would further test transferability; our current pinned ARC2 results are therefore treated as supporting evidence rather than a reproduction claim.

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